

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY
STANDARD REFERENCE MATERIALS FOR RADIOACTIVITY MEASUREMENTS

NIST radioactivity Standard Reference Materials (SRMs) represent the national basis for accurate radioactivity measurements. Some commercial companies provide secondary radioactivity standards, both for specific and general needs. These secondary standards are linked to the NIST-produced national standards through intercomparison or traceability testing programs with NIST. Generally, the SRMs that are available from NIST are provided for one of several reasons: (a) they are not available from outside commercial suppliers, (b) commercially produced standards may not be traceable to national standards, (c) the accuracy of commercially produced standards is not adequate for a significant number of users, or (d) there are enough requests from the user community for a standard not otherwise available. SRMs for radioactivity measurements can typically be classified into three general categories: (1) environment and nuclear power, (2) medicine, and (3) basic and applied research using or involving radioactivity in the development of nuclear data and the examination of basic nuclear processes.

Since the beginning of the nuclear age, the importance of accurate radioactivity measurements has been recognized, especially in the development of nuclear power and nuclear medicine. Until the 1940s, the primary concern of physicists was the establishment of the physics and chemistry of nuclear processes. Therefore, the need for higher accuracy in radioactivity measurements became more urgent. Although some rudimentary metrology, as we now think of it, had been performed before the late 1940s, the development of a metrology-based discipline began around 1950.

The coincidence counting technique, although dating back decades, was developed into a primary measurement method by taking advantage of improvements in electronics and counting system design. The coincidence counting method became the cornerstone of the international measurement system for radioactivity, to which virtually all other indirect methods were referred. During the 1950s, the basic calibration factors for many radionuclides were established and retained on a stable and accurate ionization chamber at NIST (then NBS). With this set of definitive calibration factors, it was possible to begin assessing the accuracy of user measurements in commerce and research. This assessment was first done primarily in nuclear medicine and later moved into the nuclear power and environmental areas.

The NIST Radioactivity Group conducts various programs in standards development, calibration services, traceability testing, and basic research to meet the requirements for new standards in radionuclide metrology, environmental monitoring, and nuclear medicine. Many research efforts have led to the development of unique radioactivity standards and measurements, which are transferred to industry via SRMs, calibration services, comparative measurements, and traceability testing programs.

Collaborations

Collaborations between NIST and the nuclear communities have resulted in many NIST traceability testing and measurement assurance programs. Participants include federal regulatory agencies, radiopharmaceutical manufacturers, commercial suppliers of calibration sources and services, and the nuclear power industry. One of these collaborations is the NIST Radiochemistry Intercomparison Program

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(NRIP). In NRIP, participants receive aqueous solutions, glass fiber filters, artificial urine and fecal materials, and soil that have been certified for strontium-90, uranium-234, uranium-238, plutonium-238, and americium-241 at levels as low as 0.03 Bq per sample.

The NIST natural matrix SRMs result from a collaboration of national and international environmental measurement laboratories. These natural matrix SRMs are distributed as ground, homogenized powders of soils, sediments, and organic materials and are characterized for as many as 20 radionuclides at environmental levels.

NIST Radioactive Standard Reference Materials Program

NIST issues a wide array of SRMs for radioactivity measurements. Typically, there are 60 such SRMs in stock, and calibrations have been performed on approximately 80 radionuclides. Uncertainties of these calibrations are typically 2 percent or less. These calibrations are performed using approximately 40 radiometric and a few mass spectrometric methods. The SRMs are issued in acidic or basic solutions of alpha, beta, gamma, and x-ray emitting radionuclides, and natural matrix materials. NIST radioactivity measurements (SRMs, calibrations, etc.) are intercompared with the primary standards of other national metrology laboratories worldwide through the International Bureau of Weights and Measures (BIPM), which organizes and analyzes the measurements. The Radioactivity Group, part of the NIST Physical Measurements Laboratory, prepares the radioactivity SRMs listed in this website. Radioactive SRMs are reissued periodically and may not always be available. If a NIST SRM of a particular radionuclide is not available, the following options should be considered.

Other Government Laboratories

Thorium, uranium, and plutonium-powdered ores, metal samples, and isotopic standards for mass spectrometry are available from the New Brunswick Laboratory, U.S. Department of Energy. For technical information, email richard.essex@nnsa.doe.gov. For general pricing and availability, email NBLSales@nnsa.doe.gov. The NBL internet address is [NBL Program Office | Department of Energy](#).

The national standards laboratories of other nations offer reference materials of some radionuclides based upon calibrations that have been shown to be in satisfactory agreement with those of NIST through international intercomparisons. For additional information, contact the NIST Radioactivity Group. Call 1-301-975-4338 or email radsrms@nist.gov.

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Commercial Standards Laboratories

Several firms in the U.S. prepare and distribute radioactivity reference materials. Prospective users of such products should ask suppliers if they participate in a traceability-testing program with NIST to ensure that the suppliers' calibrations are traceable to the U.S. national standards.

Calibration Services

The Radioactivity Group at NIST offers calibration services for over 80 radionuclides. However, there are limitations on the physical and chemical forms and the activity of sources that may be submitted for calibration. A list of the calibration services available from the Radioactivity Group is available at

https://shop.nist.gov/ccrz__ProductList?categoryId=a0lt00000013aDoAAI&cclcl=en_US.

Special Development Projects

For special cases where unique source requirements exist and direct traceability to NIST is required, the NIST Radioactivity Group may be able to perform research to develop and produce a bespoke reference source or material on a cost-recovery basis. Such projects are undertaken subject to resource availability and only when a commercial source is unavailable or impractical. For additional information, contact the NIST Radioactivity Group.

Additional SRM Technical Information

The amount of a radionuclide in an SRM at a specified time is stated as

- (1) the number of atoms, or
- (2) the activity (the average number of nuclear transformations per second) or
- (3) the emission rate of a particular radiation,

depending upon the method of calibration or the intended use. For solution standards, the quantity is usually specified per gram of liquid.

If the half-life is known, the number of atoms (or the mass) can be calculated from the activity and vice versa. If the fraction of radioactive decays that give rise to a specific radiation is known, the emission rate for that radiation can be calculated from the activity and vice versa. Suggested values of probability per decay for the more intense radiations usually accompany each SRM or can be obtained from the NIST Radioactivity Group.

The old, traditional unit of activity is the curie (Ci), which is defined as 3.7×10^{10} nuclear transformations per second. By international agreement, the SI unit of activity is now the becquerel (Bq), which is defined as one nuclear transformation per second. Hence, 1 curie is equal to 3.7×10^{10} becquerels. NIST reports all measures of activity using the becquerel.

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For additional technical information concerning an SRM in Category 205, contact the NIST Radioactivity Group.

NIST Uncertainty Statements

A measurement result is complete only when accompanied by a quantitative statement of its uncertainty. The uncertainty often determines the usefulness of the measurement result. Discussions among representatives of eleven national standards laboratories, following the distribution by the Bureau International des Poids et Mesures (BIPM) of a questionnaire about uncertainty statements, led to BIPM Recommendation INC-1 (1980). The assertion is that the combined uncertainty of a measurement result is best characterized by the sum of the variances (or approximations thereof) that describe the components. An expanded uncertainty is formed by multiplying the combined uncertainty by a coverage factor. (In accordance with NIST policy, a coverage factor of 2 is now used in all Radioactivity SRM Certificates).

This method of expressing uncertainties has been used for international intercomparisons in several fields without difficulty and has been implemented by all national standards laboratories. The publications below provide useful guidance, including definitions of terms and examples of calculations, when implementing this method.

- [1] JCGM 100:2008; Guide to the Expression of Uncertainty in Measurements; (ISO GUM 1995 with minor corrections), Joint Committee for Guides in Metrology: BIPM, Sevres Cedex, France (2008); available at http://www.bipm.org/utils/common/documents/jcgm/JCGM_100_2008_E.pdf
- [2] B.N. Taylor and C.E. Kuyatt, Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results, NIST Technical Note 1297, 1994. Available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20407. U.S.A. Single copies may be requested from NIST by sending a fax to 1-301-926-2884.

License Certification

U.S. Government regulations require all suppliers of radioactive material to verify, prior to shipment, that the purchaser is authorized to receive the type, form, and quantity of radioactive material that is to be transferred. For some SRMs, indicated in the NRC License/Certification or Equivalent Required column: <https://tsapps.nist.gov/srmext/tables/205.4.pdf>, NIST requires that all transfers shall have documentation that includes a current copy of the recipient's license issued by the NRC or an Agreement State. In addition, a certification from the recipient indicating that they are authorized to possess the type, form, and quantity of the SRM to be transferred is required. SRM purchasers should check any other regulatory requirements, if applicable, prior to acquiring the material.